

IFT-SafePredict — Manual Test Guide

This guide walks through every feature. Run the dev server first:

```
cd ift-safepredict
yarn dev
```

Open the URL shown in terminal (typically <http://localhost:5173/>).

1. Core Prediction

- ☐ **Brine required:** Leave brine type blank → Predict button stays greyed out, error text in red
- ☐ **Predict IFT:** Select **NaCl**, enter P=15, T=350, keep defaults → click Predict IFT
- ☐ **Results appear:** Three-column P10 / P50 / P90 with values in mN/m
- ☐ **Regime badge:** Shows "Supercritical" (blue) or "Subcritical" (purple) in the input card header
- ☐ **QA badge:** Shows GREEN/YELLOW/RED with message below results
- ☐ **EOS tag:** " $\Delta\rho$ EOS-estimated" badge appears in results since ρ_{rho} auto-computed
- ☐ **Pr/Tr display:** Shows at bottom of results card

2. Input History (IndexedDB)

- ☐ **Auto-save:** After predicting, expand the "Input History" section below the chart — shows the prediction you just ran
- ☐ **Reload inputs:** Click any history entry → inputs repopulate; click Predict again
- ☐ **Delete:** Click the X button on any history entry → it disappears
- ☐ **Clear all:** Click "Clear All History" → list empties
- ☐ **Persistence:** Refresh the page → expand Input History → previous entries still there

3. Shareable URLs

- ☐ **URL updates:** After predicting, the browser URL ends with **#P=15&T=350&brine=NaCl...**
- ☐ **Share button:** In Results card, click "🔗 Share" → "✓ Copied" flashes
- ☐ **Open shared link:** Copy the full URL, paste in a new tab → inputs auto-populate and auto-predict
- ☐ **Cross-device:** Send the link to a colleague (or different browser) → same result

4. Reservoir Presets

- ☐ **Preset dropdown:** At top of input card, select "Sleipner (Utsira Fm, North Sea)" → all inputs populate
- ☐ **Predict from preset:** Click Predict → realistic result for Sleipner conditions
- ☐ **All 13 presets work:** Try a few: Snøhvit, Quest, Gorgon, Pure Water → each gives reasonable IFT

- ☐ **Preset clears selection:** After selecting a preset, dropdown resets to "Select a CCUS formation..."

5. Depth Converter

- ☐ **Expand:** Click "Depth → Pressure & Temperature converter" in input card
- ☐ **Default values:** Shows depth=1500m, 25°C/km gradient, 10 MPa/km gradient
- ☐ **Preview:** Shows $T \approx 52.5^\circ\text{C}$ (325.6 K) and $P \approx 15.00$ MPa
- ☐ **Apply:** Click "Apply P & T" → P and T input fields update
- ☐ **Custom values:** Change depth to 3000m → shows $T \approx 90^\circ\text{C}$ → Apply → fields update again
- ☐ **Unit-aware:** Switch units to "Field (bar, °C)" → depth preview shows °C and bar correctly

6. Sensitivity Analysis (Tornado Chart)

- ☐ **Shows after predict:** Below Uncertainty Chart, "Sensitivity Analysis" card appears
- ☐ **Baseline IFT:** Shows "Baseline IFT = XX.XX mN/m"
- ☐ **Tornado bars:** Horizontal bars showing ΔIFT when each parameter varies min→max
- ☐ **Sorted by impact:** Most impactful parameter at top
- ☐ **Color coding:** Blue = decreases IFT, Red = increases IFT
- ☐ **Resolution selector:** Change from 5 to 10 steps → bars update noticeably
- ☐ **Re-predict:** Change inputs and re-predict → sensitivity chart updates

7. P–T Contour Map

- ☐ **Tab:** Click "P–T Map" tab
- ☐ **Heatmap renders:** Colored grid shown with P on x-axis, T on y-axis
- ☐ **Color bar:** Right side shows IFT color scale (blue=low, red=high)
- ☐ **Regime boundary:** White dashed lines show supercritical boundary
- ☐ **Critical point:** White dot with "Critical point" label near bottom-left
- ☐ **Hover tooltip:** Move mouse over any cell → tooltip shows P, T, IFT, regime
- ☐ **Click to select:** Click any cell → switches to Predict tab with that P, T loaded
- ☐ **Range controls:** Change P range to 10–30, T range to 310–360 → map refocuses
- ☐ **Inverse mode:** Check "Show safe zone" → cells where $\text{IFT} \leq \text{target}$ are highlighted white
- ☐ **Target IFT:** Change target to 20 mN/m → highlighted zone shrinks
- ☐ **Unit-aware:** Switch to "Field (bar, °C)" → axis labels and range inputs update

8. Batch Predict

- ☐ **Tab:** Click "Batch Predict" tab
- ☐ **Upload prompt:** Shows upload area with CSV requirements
- ☐ **Test CSV:** Create a file `test_batch.csv` with:

```
P,T,salinity,MCM,BCM,x_CH4,x_N2,brineType
10,320,1.0,1.0,0,0,0,NaCl
15,330,2.0,2.0,0.5,0.05,0.02,NaCl+CaCl2
25,370,3.5,3.0,0.5,0.1,0.01,NaCl
5,300,0.5,0.5,0,0,0,Water
```

- ☐ **Upload:** Click " Upload CSV" → select file → shows "4 rows loaded"
- ☐ **Run batch:** Click "Run Batch Prediction (4 rows)" → progress bar fills 25%, 50%, 75%, 100%
- ☐ **Results table:** Shows 4 rows with P, T, brine, regime, P50, P10, P90, QA, UIF
- ☐ **Download CSV:** Click "↓ Download CSV" → saves CSV file with all results
- ☐ **Download JSON:** Click "↓ Download JSON" → saves JSON with all results
- ☐ **Error handling:** Test with missing required column → alert "Missing required column"

9. CO₂ Storage Capacity Estimator

- ☐ **Shows after predict:** Below Sensitivity Analysis in Predict tab
- ☐ **Auto-calculates:** Trapping Efficiency derived from IFT using power-law model
- ☐ **Stat cards:** Shows IFT, Trapping Efficiency %, Storage Capacity (Mt CO₂)
- ☐ **Adjust inputs:** Change area to 500 km², thickness to 100m → storage capacity updates
- ☐ **Scenario comparison:** Lower IFT → higher trapping efficiency → more storage
- ☐ **Summary text:** Reads "A 500 km² × 100m reservoir could store ~X Mt CO₂"

10. Light/Dark Theme

- ☐ **Toggle:** Click  button in header → page switches to light theme immediately
- ☐ **Light appearance:** White cards, dark text, softer borders — readable in bright room
- ☐ **Toggle back:** Click  → returns to dark theme
- ☐ **Persistence:** Refresh the page → theme stays as last selected
- ☐ **All components themed:** Check that input fields, cards, charts, tables all look correct in both themes
- ☐ **No flash:** On reload, page loads in correct theme instantly (no white flash)

11. Unit System

- ☐ **Default:** Starts in "SI (MPa, K)"
- ☐ **Switch to Field:** Select "Field (bar, °C)" → P shows in bar, T shows in °C
- ☐ **Predict in Field units:** Set P=150 bar, T=50°C → predict → converts internally
- ☐ **Switch to US:** Select "US (psi, °F)" → P in psi, T in °F
- ☐ **Verify conversion:** 15 MPa → ~2175 psi, 323 K → ~122°F
- ☐ **Unit persistence:** Refresh → unit system stays
- ☐ **Contour Map units:** P–T Map tab axes and range inputs match current unit system
- ☐ **Depth Converter:** Preview values in the current unit system

12. Keyboard Shortcuts

- ☐ **Ctrl+Enter:** While on Predict tab, press Ctrl+Enter → triggers prediction
- ☐ **Cmd+Enter (Mac):** Same as Ctrl+Enter on macOS
- ☐ **Escape:** Press Escape → all inputs reset to defaults, result clears
- ☐ **Shortcut hint:** "Ctrl+Enter to predict · Escape to reset" shown below Predict button

13. Input Validation

- ☐ **P out of domain:** Set P=70 → ⚠ icon on Pressure field, "Domain: 0.1–60 MPa" hint
- ☐ **T out of domain:** Set T=550K (or 277°C) → ⚠ on Temperature field
- ☐ **Yellow border:** Out-of-domain fields get yellow border
- ☐ **Still predicts:** Prediction still runs (QA shows RED for extrapolation)
- ☐ **Unit-aware:** Switch to bar → domain shows in bar (1–600 bar)

14. Export Features

- ☐ **CSV export:** After predicting, click "↓ CSV" → downloads **.CSV** with inputs + results
- ☐ **PDF export:** Click "↓ PDF" → downloads formatted PDF report

15. Uncertainty Chart

- ☐ **Shows after first prediction:** Prediction History chart renders below results
- ☐ **Updates:** Run 3+ predictions with different values → chart shows multiple lines
- ☐ **Uncertainty band:** Shaded area between P10–P90 visible
- ☐ **Color by QA:** P50 line color matches GREEN/YELLOW/RED status

16. Scenario Comparison

- ☐ **Save scenarios:** After predicting, click "+ Save to comparison" → "Scenario 1" saved
- ☐ **Save up to 3:** Save 3 scenarios → button shows "(max 3)" and disables
- ☐ **Compare tab:** Switch to "Compare Scenarios" → shows 3 scenario cards side-by-side
- ☐ **Remove:** Click X on any scenario → removed
- ☐ **Export comparison:** "↓ Export CSV" → downloads CSV with all scenarios

17. Equations Tab

- ☐ **Renders LaTeX:** Click "Equations" → 7 sections of properly rendered equations
- ☐ **Scrolling:** Tables and equations scrollable horizontally on narrow screens

18. About Tab

- ☐ **Model info:** Shows model versions, nRMSE values
- ☐ **Citation:** Shows paper citation ready to copy
- ☐ **Privacy notice:** States "100% client-side"

19. Analytics Dashboard (admin only, requires Google Sheets setup)

The Analytics page is hidden from the tab bar. Access it via the **"admin"** link in the page footer.

- ☐ **Hidden link:** Scroll to footer → click "admin" link (subtle, low-contrast)
- ☐ **Password gate:** Password input shown when accessed
- ☐ **Wrong password:** Enter wrong password → "Unauthorized" error
- ☐ **Correct password:** Enter password from **tracking.js** → dashboard loads
- ☐ **Summary cards:** Total visits, Unique IPs, Today, Avg Session shown
- ☐ **Charts:** Daily page views line chart + Device doughnut chart render
- ☐ **Tables:** Top Pages, Countries, Recent Visitors tables populated

- ☐ **No backend message:** If `APPS_SCRIPT_URL` is empty → setup instructions shown
-

Regression Tests

Before deploying, run the automated suite:

```
yarn test
```

Expected output: `Tests: 15 passed`

And verify the build still works:

```
yarn build
```

Expected: ✓ `built in X.XXs` with no errors.